Calculus II Lesson 27 Conics Sections - Hyperbola

1) $\frac{x^2}{4} - \frac{(y+2)^2}{25} = 1$

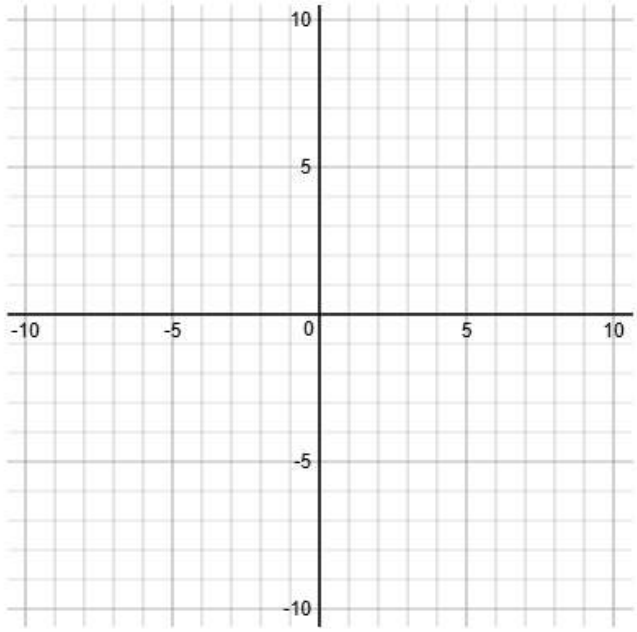
a) Find the center of the hyperbola:_____

b) Find the vertices:_____

c) Find the foci:_____

d) Graph $\frac{x^2}{4} - \frac{(y+2)^2}{25} = 1$

Hint: Use the online 2D Calculator.



2) $\frac{(x-1)^2}{25} - \frac{(y+2)^2}{4} = 1$

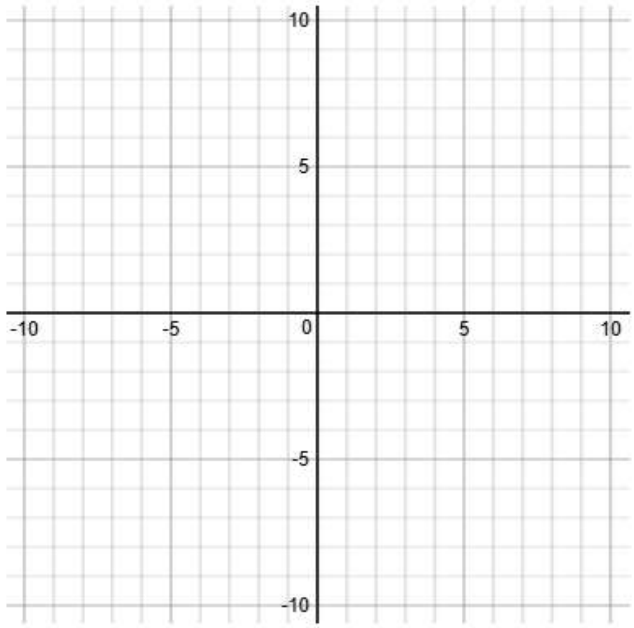
a) Find the center of the hyperbola:_____

b) Find the vertices:_____

c) Find the foci:_____

d) Graph $\frac{(x-1)^2}{25} - \frac{(y+2)^2}{4} = 1$

Hint: Use the online 2D Calculator.



3) $4x^2 - 9y^2 + 24x - 36y + 12 = 0$

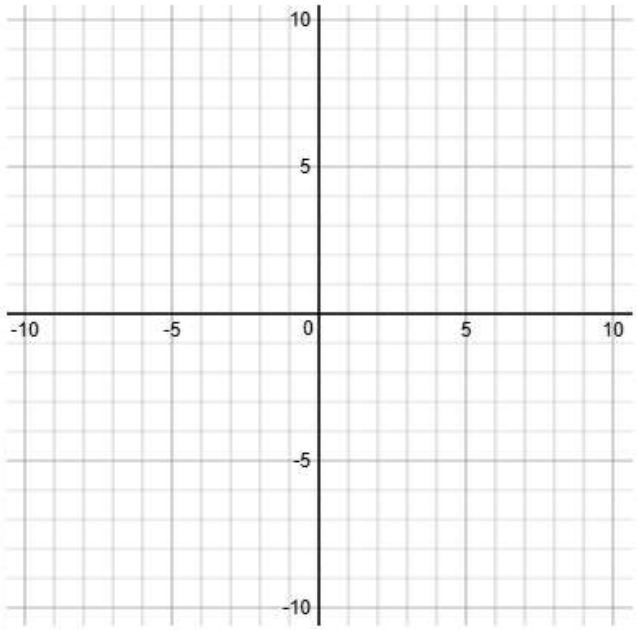
a) Find the center of the hyperbola:_____

b) Find the vertices:_____

c) Find the foci:_____

d) Graph $4x^2 - 9y^2 + 24x - 36y + 12 = 0$

Hint: Use the online 2D Calculator.



4) Find an equation of the hyperbola that has the following:

Vertices: (0, 8) and (0, 2); Foci: (0,10), (0,0)

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) $b^2 =$ _____ Note: For hyperbola, $a^2 + b^2 = c^2$

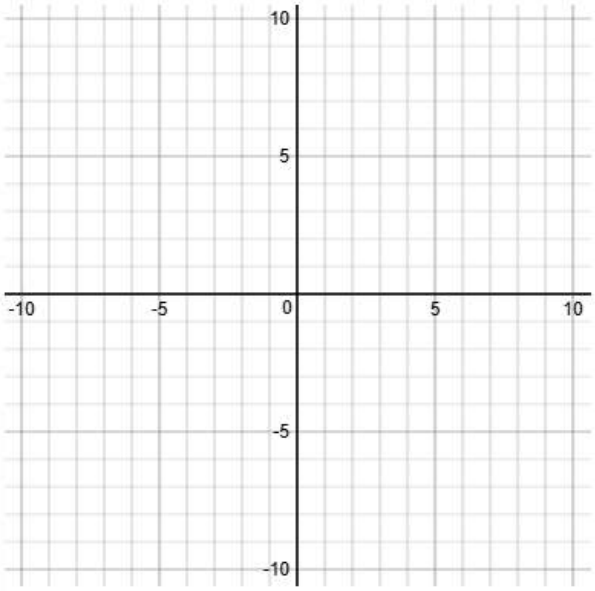
d) Equation of hyperbola:_____multiple choice_____

A) $\frac{(y-5)^2}{16} - \frac{(x)^2}{9} = 1$ B) $\frac{(y-5)^2}{9} - \frac{(x)^2}{16} = 1$

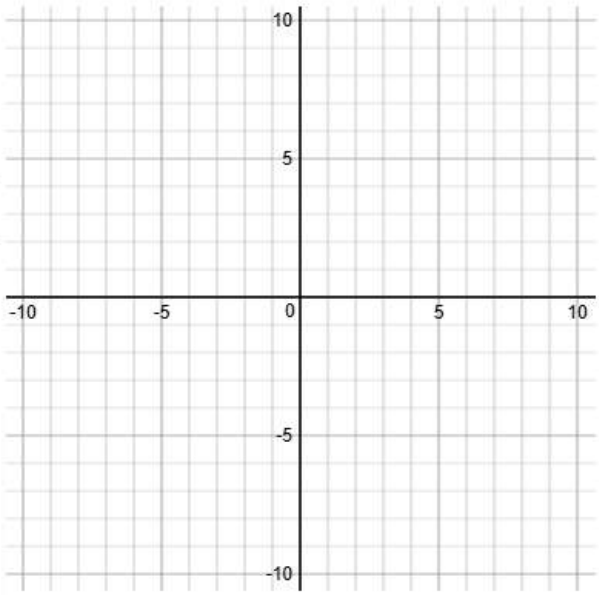
C) $\frac{(y-5)^2}{9} - \frac{(x-2)^2}{16} = 1$ D) $\frac{(y-5)^2}{16} - \frac{(x-2)^2}{9} = 1$

e) Graph equation of hyperbola.

Hint: Use the online 2D Calculator.



5) Find an equation of the hyperbola that has the following:
Center:(0, 0) Vertex: (3, 0) ; Focus(4, 0)



a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) b^2 = _____ Note: For hyperbola, $a^2 + b^2 = c^2$

d) Equation of hyperbola:_____

A) $\frac{x^2}{7} - \frac{y^2}{9} = 1$ B) $\frac{x^2}{9} - \frac{y^2}{7} = 1$

C) $\frac{x^2}{9} - \frac{y^2}{4} = 1$ D) $\frac{x^2}{4} - \frac{y^2}{9} = 1$

e) Graph equation of hyperbola.

Hint: Use the online 2D Calculator.

6) Find an equation of the hyperbola that has the following:
Center:(0, 0) Vertex: (0, 6) ; Focus(0, 10)

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) b^2 = _____ Note: For hyperbola, $a^2 + b^2 = c^2$

d) Equation of hyperbola:_____

7) Find an equation of the hyperbola that has the following:
Vertices: (0, 0) and (0, 10); Foci: (0,-2), (0,12)

a) a = distance from center to vertex = _____

b) c = distance from center to focus = _____

c) b^2 = _____ Note: For hyperbola, $a^2 + b^2 = c^2$

d) Equation of hyperbola:_____